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# Hiroki SAWADA

## Education

2016.4 – 2020.3 **Keio University**
BA in Applied Physics and Physico-Informatics
Kanagawa, Japan

2020.9 – 2026.6 **Okinawa Institute of Science and Technology Graduate University**
Ph.D. in Computer Science – Cognitive Neuro-robotics Research Unit
Okinawa, Japan

## Research Experience

2026.4 – present **Sony Computer Science Laboratory, Humanity Science Project**
Project Researcher
Tokyo, Japan

**Body Integration Learning Framework for Highly Skilled Robots**

- Leading the setup of physical human-robot interaction experiments using the open-source humanoid robot “OpenArm” (Enactic).
- Proposed PredVLA, a compact recurrent controller organized around predicting sensorimotor dynamics and inferring its latent state from prediction error, rather than compressing a conventional observation-to-action policy [1].
- Proposed LIBERO-CTRL, a six-axis benchmark that pairs single-axis and compound perturbations from identical initial states, revealing compound-robustness failures of VLA policies that aggregate success rates hide [2].

2025.1 – 2025.7 **Équipes Traitement de l’Information et Systèmes Lab, CY Cergy Paris University**
Prof. Alexandre Pitti, Prof. Mathias Quoy
Cergy, France

**Investigation of Robust Predictive-Coding inspired Recurrent Neural Network in Real Robot Tasks**

- Proposed CERNet, a hierarchical predictive-coding RNN with class embedding, enabling unified real-time motion generation, recognition, and meta-inference of confidence.
- Demonstrated stable reproduction of 26 handwritten alphabet trajectories and confidence estimation using the Reachy humanoid robot (Pollen Robotics) in both simulation and real-world settings.
- Published in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2026* [7].

2021.9 – 2026.6 **Cognitive Neuro-robotics Research Unit, Okinawa Institute of Science and Technology Graduate University**
Prof. Jun Tani
Okinawa, Japan

**Human-Robot Kinaesthetic Interaction Based on the Free Energy Principle**

- Investigated human-robot kinaesthetic interaction using a predictive coding-inspired variational RNN, guided by the Free Energy Principle, to model bidirectional sensorimotor dynamics between humans and a humanoid robot.

- Through mutually adaptive physical behaviors through active inference, conducted research on revealing the relationship between kinaesthetic perception and the internal state of the employed variational RNN [5].
- Studied the emergence of novel behaviours through incremental learning with variational RNNs [6].
- Proposed a framework where a network can continuously learn new movements through human-robot interaction by gating the learning rate based on the free energy [4].
- Co-developed a mechanism that switches PV-RNN meta-priors online according to accumulated reconstruction error, letting the robot regulate its reliance on internally generated predictions versus external sensory evidence [3].

2019.7 – 2020.1
2021.1 – 2021.4

**Quantum Materials Science Unit, Okinawa Institute of Science and Technology Graduate University**
Prof. Yoshinori Okada
Okinawa, Japan

**Evaluation of magnetic properties in van der Waals compounds**

- Evaluated the magnetic properties of van der Waals compounds $CeTe_3$ and $CeTe_2Se$ (collaborated with Dr. Imai and Dr. Fujisawa). Results published as a full journal paper [8].

2019.4 – 2020.3

**Matoba & Kamihara Lab, Keio University**
Prof. Masanori Matoba, Prof. Yoichi Kamihara
Kanagawa, Japan

**Synthesis of layered oxysulfide to optimize thermoelectric properties**

- Synthesised polycrystalline $LaCuS_{0.5}Se_{0.5}O$, which is expected to have high thermoelectric performance, for the first time.
- Evaluated the effect of copper deficiency in the polycrystalline sample on its thermoelectric properties. Results published in full journal papers [9, 10].

## Working Experience

2026.3 – 2026.4

**Mercor**
Bilingual Japanese STEM Expert Writer & Editor
Tokyo, Japan

**Designing scientific prompts and evaluation frameworks for AI systems**

- Designed Prompt-Rubrics for evaluating LLMs in the fields of science and engineering.
- Worked as both a prompt creator (Writer) and an evaluator (Editor).

2023.9 – 2026.3

**Reazon Holdings, Inc.**
R&D Engineer
Tokyo, Japan

**Development of Automatic Minutes Sender Bot Using LLM and NLP**

- Full-time software engineer (4 days/week) with permanent employment status.
- Lead engineer for an automated meeting-minutes distribution system that integrates LLMs and NLP; authored over 95% of the code. The system is now running in production within the company.
- Built a scalable backend leveraging AWS EC2, S3, SageMaker Asynchronous Inference, Lambda, and other services to enable asynchronous inference.
- Designed and developed a RAG-based GPT agent capable of answering complex queries regarding internal company policies and regulations, improving internal knowledge accessibility and employee self-service efficiency.

## Publications and Presentations (arXiv preprint, under review)

[1] **Hiroki Sawada** and Shunichi Kasahara. “PredVLA: Predictive Sensorimotor Modeling for Sub-Million-Parameter Robot Manipulation.” *arXiv preprint arXiv:2608.26673* (2026). (Under Review)

[2] **Hiroki Sawada** and Shunichi Kasahara. “Beyond Single-Axis Testing: Paired Evaluation of Compound Robustness in Vision-Language-Action Policies.” *arXiv preprint arXiv:2609.15940* (2026). (Under Review)

[3] Henrique Oyama, **Hiroki Sawada**, and Jun Tani. “Predictive-Coding-Based Autonomous Regulation of Internally Generated and Externally Coupled Processing in Human-Robot Interaction.” *arXiv preprint arXiv:2609.06888* (2026). (Under Review)

[4] **Hiroki Sawada** and Jun Tani. “Free-Energy-Gated Plasticity for Real-Time Online Motor Learning in Physical Human-Robot Interaction.” *arXiv preprint arXiv:2608.23000* (2026). (Under Review)

## Publications and Presentations (peer-reviewed)

[5] **Hiroki Sawada**, Wataru Ohata, and Jun Tani. “Human-Robot Kinaesthetic Interaction Based on The Free Energy Principle.” *IEEE Transactions on Systems, Man, and Cybernetics: Systems* (2024).

[6] **Hiroki Sawada** and Jun Tani. “Incremental Learning in Physical Human-Robot Interaction Using Predictive-Coding inspired Variational RNNs.” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Late-Breaking Poster* (2024).

[7] **Hiroki Sawada**, Alexandre Pitti, and Mathias Quoy. “CERNet: Class-Embedding Predictive-Coding RNN for Unified Robot Motion, Recognition, and Confidence Estimation.” *IEEE International Conference on Robotics and Automation (ICRA), pp. 21744–21751* (2026).

[8] Daichi Ueta, Riki Kobayashi, **Hiroki Sawada**, et al. “Anomalous Magnetic Moment Direction under Magnetic Anisotropy Originated from Crystalline Electric Field in van der Waals Compounds $CeTe_3$ and $CeTe_2Se$.” *Journal of the Physical Society of Japan 91(9): 094706* (2022).

[9] Nobuhiko Azuma, **Hiroki Sawada**, Houji Itoh, et al. “Thermal transport properties of a mixed anion layered compound, polycrystalline $LaCu_{1-\delta}S_{0.5}Se_{0.5}O$ ($\delta = 0.01$).” *Korean Journal of Materials Research 34, 464–474* (2024).

[10] Nobuhiko Azuma, **Hiroki Sawada**, Houji Itoh, et al. “In-gap-states of a mixed anion layered compound, polycrystalline $LaCu_{1-\delta}S_{0.5}Se_{0.5}O$ ($\delta$~0.01) as a degenerate semiconductor.” *Materials Science and Technology Japan 58, 64–68* (2021).

## Personal Projects

2026.1

**Magic Tower Revival (魔法の塔 復刻版) — iOS app, available on the App Store**
Personal Project | C#, Unity, iOS | App Store

- Remade “Magic Tower,” a 1995 roguelike dungeon RPG released by N.W for the PC-9800 series, for iPhone with permission from the original author.
- Faithfully reproduced the original 50-floor layouts, enemy stats, and event sequences, with a UI redesigned for modern smartphones. Built with C# and Unity.
- Solely responsible for every stage, from planning, UI design, and development to App Store release and operation (free to play with one-time in-app purchases).

2025.3 – present

**LLM-based Mobile Game App (in development)**
Personal Project | Swift, Python, OpenAI API

- Designing and developing an iOS game app where users ask questions to guess an answer the AI has thought of.
- Integrated OpenAI GPT API and AWS Lambda for real-time inference and data handling.
- Applied prompt engineering techniques to constrain and guide model responses across different genres (e.g., animals, brands, countries).

- Leveraged Retrieval-Augmented Generation (RAG) to reduce API call cost while maintaining consistency and relevance.
- Built entirely using Swift and Xcode as a native iOS app. Demo available upon request.

## Awards and Honours

2020.9 – 2026.6 **Fully Funded Ph.D.**
Okinawa Institute of Science and Technology Graduate University, Okinawa, Japan
- Selected for a fully funded Ph.D. program.

2021.4 – 2022.3 **JEES Softbank AI Researcher Fellowship**
Softbank, Japan
- Scholarship of 1,000,000 yen without usage limitations.
- Granted to 100 Ph.D. students in the field of AI.

2025.1 – 2025.7 **Bourses France Excellence 2025**
French Government, France
- Scholarship from the French government to Japanese PhD students.
- Covers full tuition and a monthly stipend of €1709 for 6 months.

2025.9 **TOEIC Listening & Reading test: score 975 / 990**

## Language

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|---|---|
| English | Near native-level fluency *(Lived 13 years abroad in an international environment; PhD program in English)* |
| Japanese | Native |
| French | Conversational proficiency (CEFR A2 level) |

## Activities

2022.2 – 2023.2 **Diversity Officer, OIST Student Council**
- Promoted interaction among students from diverse cultural backgrounds within OIST.
- Organized a campus symposium “What is discrimination?”.

2023.2 – 2024.2 **Culture and External Relations Representative, OIST Student Council**
- Strengthened relationships between OIST and the local community.
- Co-hosted two sake introduction events (≈50 participants each) in collaboration with Miss Sake, an organization promoting Japanese sake worldwide.
- Organized a joint cultural event with Okinawa Prefectural University of Arts.

2026.1 **Reviewer, IEEE ICRA 2027**
- Reviewed main conference proceedings papers for the IEEE International Conference on Robotics and Automation (ICRA) 2027.

2025.5 **Reviewer, IEEE ICDL 2025**
- Reviewed peer-reviewed contributed papers in the field of cognitive robotics and learning for the IEEE International Conference on Development and Learning (ICDL) 2025.